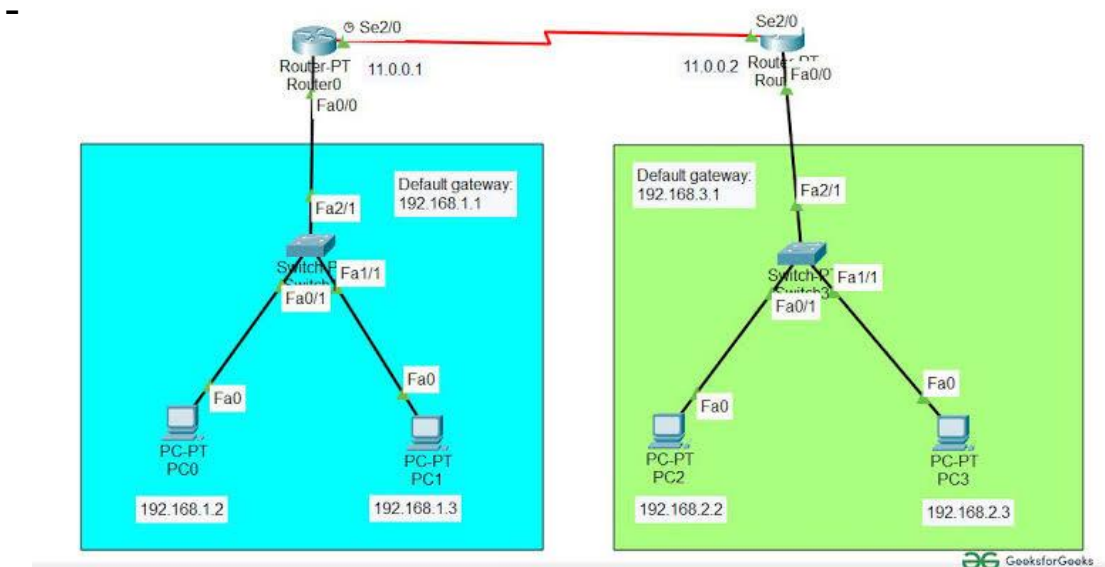


Day – 1 Static router concept theory

Task – 1 Set up a simple network in cisco tracer.



Task – 2 Make a router 1 static to the pc and router 2 to lan.

- Network Addressing Scheme
 - Router2 LAN Network: 192.168.2.0/24
 - Router2 Gigabit Ethernet Interface (Gi0/0): 192.168.2.1
 - PC2 (Connected to Router2's LAN Switch):
 - o IP Address: 192.168.2.10
 - o Subnet Mask: 255.255.255.0
 - o Default Gateway: 192.168.2.1
- 1.Configure Static Route on Router1:Routing Table Update. In the CLI of Router1, define a route targeting Router2's LAN using Router2's serial interface IP (10.0.0.2) as the next hop:
- Router1 ip route 192.168.2.0 255.255.255.0 10.0.0.2

- 2.Test Connectivity from Router1 to PC2:ICMP Ping Test. Open the CLI on Router1 (in privileged EXEC mode) and run:
- Router1# ping 192.168.2.10

Task – 3 Two different router in long distance.

- Zomato Network Topology Plan :-
- Ahmedabad Branch (Router1):
 - o LAN Network: 192.168.1.0/24
 - o Gateway (Gi0/0): 192.168.1.1
 - o Serial Interface (S0/1/0): 10.0.0.1/30
- Surat Branch (Router2):
 - o LAN Network: 192.168.2.0/24
 - o Gateway (Gi0/0): 192.168.2.1
 - o Serial Interface (S0/1/0): 10.0.0.2/30
- WAN Link: 10.0.0.0/30
- 1.Configure Router1 (Ahmedabad) LAN & WAN Interfaces:Ahmedabad Gateway.
- Open the CLI tab on Router1 and configure the local branch gateway and WAN point-to-point IP:
- Router> enable
 Router# configure terminal
 Router(config)# hostname Ahmedabad-R1
 Ahmedabad-R1(config)# interface GigabitEthernet0/0
 Ahmedabad-R1(config-if)# ip address 192.168.1.1 255.255.255.0
 Ahmedabad-R1(config-if)# no shutdown
 Ahmedabad-R1(config-if)# exit

 Ahmedabad-R1(config)# interface Serial0/1/0
 Ahmedabad-R1(config-if)# ip address 10.0.0.1 255.255.255.252
 Ahmedabad-R1(config-if)# clock rate 64000
 Ahmedabad-R1(config-if)# no shutdown
 Ahmedabad-R1(config-if)# exit

- 2.Configure Router2 (Surat) LAN & WAN Interfaces:Surat Gateway.
- Open the CLI tab on Router2 and configure the local branch gateway and WAN point-to-point IP:
- Router> enable
Router# configure terminal
Router(config)# hostname Surat-R2
Surat-R2(config)# interface GigabitEthernet0/0
Surat-R2(config-if)# ip address 192.168.2.1 255.255.255.0
Surat-R2(config-if)# no shutdown
Surat-R2(config-if)# exit

- Surat-R2(config)# interface Serial0/1/0
Surat-R2(config-if)# ip address 10.0.0.2 255.255.255.252
Surat-R2(config-if)# no shutdown
Surat-R2(config-if)# exit

- 3.Configure Static Route on Ahmedabad Router:Outbound Route.
- Tell the Ahmedabad router how to reach the Surat LAN (192.168.2.0/24) via Surat's serial IP (10.0.0.2):
- Ahmedabad-R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2

- 4.Configure Static Route on Surat Router:Inbound Route.
- Tell the Surat router how to reach the Ahmedabad LAN (192.168.1.0/24) via Ahmedabad's serial IP (10.0.0.1):
- Surat-R2(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1

- 5.Configure Branch Client PCs:Host IP Setup.
- Ahmedabad PC (Terminal 1):
 - o IP Address: 192.168.1.10
 - o Subnet Mask: 255.255.255.0
 - o Default Gateway: 192.168.1.1
- Surat PC (Terminal 2):
 - o IP Address: 192.168.2.10
 - o Subnet Mask: 255.255.255.0
 - o Default Gateway: 192.168.2.1
 - o
- 6.Test End-to-End Connectivity:Inter-Branch Ping.

- Open the Command Prompt on the Ahmedabad PC (192.168.1.10) and ping the Surat PC (192.168.2.10):
- DOS
- C:\> ping 192.168.2.10

Task – 4 Advantages and Disadvantages.

- Advantages of Static Routing :-
 - 1. Zero Control-Plane Overhead & Maximum Hardware Performance :- Static routing does not require routers to run route-calculation algorithms (like Dijkstra's or Bellman-Ford) or continuously broadcast routing updates across the network. This frees up 100% of the router's CPU and RAM for packet forwarding. Example (IRCTC Tatkal Booking Hours): During high-concurrency peak windows—such as daily Tatkal booking at 10:00 AM—IRCTC database servers and edge switches face millions of requests per second. Using static routes between core database servers and static API gateways ensures zero router processing power is wasted on route calculation, allowing every megahertz of CPU to handle passenger transactions without queue drop or routing delays.
 - 2. Deterministic Pathing & Enhanced Security :- Because routes are hardcoded manually by network engineers, traffic follows an exact, predictable path. Routers will not accept unauthorized or spoofed dynamic route advertisements from untrusted devices, reducing the attack surface. Example (Swiggy Payment Gateway Processing): When a user places an order on Swiggy and processes a payment via UPI or credit card, the transaction request is routed over a predetermined, hardcoded static path directly to the bank's secure payment switch. This guarantees that sensitive financial data cannot be rerouted through an unverified or intercepted path caused by dynamic routing table manipulation or malicious route injection.
- Disadvantages of Static Routing :-
 - 1. Lack of Automatic Failover (Single Point of Failure) :- Static routes cannot detect when an intermediate physical link or network node goes down. If a connection fails, the router continues sending traffic into the "black hole" until a network administrator manually logs in and updates the route. App Example (Swiggy Live Order & Delivery Partner Tracking): If the primary network link between Swiggy's regional dispatch server and a city's driver location hub cuts out during peak dinner

hours, a static route cannot adapt. As a result, live GPS tracking freezes, delivery updates fail to reach the user, and driver dispatch halts completely until an engineer manually points the router to a backup link. Dynamic routing protocols (like OSPF or BGP), by contrast, would reroute packet traffic over a redundant link within seconds.

- 2. High Administrative Overhead and Poor Scalability :- In static routing, every new network, subnet, or router addition requires manual configuration updates across every inter-connected router in the infrastructure. As the topology expands, management complexity grows exponentially. Example (IRCTC Expanding Metro & Regional Station Networks): If IRCTC expands its backend infrastructure to integrate hundreds of new local railway stations or regional ticket booking counters across India, using static routing would force network engineers to manually program static routes on every core, distribution, and edge router across the country. A single typo in an IP subnet mask could break connectivity for an entire region, making rapid infrastructure expansion prone to human error.
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